Two Architectures for Simulating Biomarker-Treatment Interactions: Implications for Statistical Power Under Carryover

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1 Abstract

Background. Simulation studies of predictive biomarker-treatment interactions in N-of-1 clinical trials require a data-generating process (DGP) that specifies how the biomarker moderates treatment response. Two fundamentally different DGP architectures are used in practice, and the choice between them determines the extent to which carryover effects reduce statistical power to detect the interaction. The published simulation literature does not distinguish them.

Methods. We formalise two architectures: direct mean moderation (Architecture~A), under which the biomarker scales the treatment effect in the population mean structure; and differential correlation (Architecture~B, the multivariate normal approach), under which the interaction emerges from treatment-state-dependent correlation between biomarker and response in the joint distribution. Power consequences under both architectures are quantified by Monte Carlo simulation on an aggregated N-of-1 design framework with continuous-time AR(1) residual correlation in the analysis model. The biomarker, treatment phase, and carryover decay are parameterised to the prazosin/PTSD reference application.

Results. Under Architecture~A, power is largely preserved under carryover because the proportional relationship between biomarker and drug response is maintained at every timepoint. Under Architecture~B, carryover erodes the differential correlation signal and reduces power. At the prazosin-calibrated reference cell ($N = 35$, $c_{bm} = 0.45$, OL+BDC design) across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, Architec-

ture~B power drops from 0.777 to 0.539 (a 30.6% relative loss; $z = 7.64$, $p < 10^{-13}$ on the B-versus-A contrast), while Architecture~A drops only from 0.751 to 0.730 (a 2.8% relative loss). The B-versus-A absolute power-loss gap of 21.7 percentage points at OL+BDC, $t_{1/2} = 1.0$ is approximately 11 times the matched Architecture~A loss; the Hybrid design produces a smaller but still significant gap ($z = 3.96$, $p < 10^{-4}$); the traditional crossover design produces no detectable gap because its within-subject AR(1) structure dominates the carryover-mediated correlation channel that Architecture~B uses. The 36-cell production factorial (2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels) was run at 1{,}000 replicates per cell with 100% convergence; MCSE on cell power is 0.013-0.016. Type~I error sits in $[0.010, 0.074]$, with no architecture-specific inflation.

Conclusions. The two architectures encode different biological mechanisms, heterogeneity at the mean structure versus heterogeneity at the variance structure, and the choice between them is a substantive scientific question rather than a parameterisation convenience. We recommend explicit declaration of the DGP architecture in any biomarker-validation simulation study and provide guidance for selecting the architecture appropriate to a given clinical context.

---

2 1. Introduction

[1] **The central question is biomarker-treatment interaction; simulation studies need a DGP that encodes it.**

- Does a baseline biomarker predict differential treatment response, i.e., does it moderate the treatment effect?
- Simulation studies are the principal power evaluation tool for proposed trial designs.
- These simulations require a DGP that explicitly encodes the biomarker-treatment interaction mechanism.

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A central question in precision medicine trial design is whether a baseline biomarker predicts differential treatment response, that is, whether the treatment effect is moderated by the biomarker value. Simulation studies are the principal tool available for evaluating the statistical power of proposed trial designs to detect such interactions, and these simulations necessarily require a data-generating process (DGP) that encodes the biomarker-treatment interaction mechanism.

[2] **Every simulation framework must choose how to generate treatment effect heterogeneity, and that choice has consequences.**

- HTE literature (2005, 2020) distinguishes qualitative interactions (direction reverses) from quantitative ones (magnitude varies).
- Any simulation framework must decide how to encode this heterogeneity in the DGP.
- That decision is substantive, not a mere parameterization detail.

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The statistical literature on treatment effect heterogeneity, developed notably by Rothwell [@rothwell2005] and synthesized more recently by Kent et al. [@kent2020path], distinguishes between *qualitative* interactions, in which the direction of the treatment effect reverses across biomarker strata, and *quantitative* interactions, in which the magnitude varies but the direction remains consistent. As we shall argue, any simulation framework intended to evaluate power for biomarker-moderated effects must choose how to generate this heterogeneity, and that choice is not merely a technical convenience.

[3] **Architecture A (direct mean moderation) is the near-universal standard; the Hendrickson (2020) MVN approach is apparently unique.**

- Every major simulation framework uses direct mean moderation: Simon (2010) for stratified designs; Freidlin and Korn (2014) for adaptive enrichment; Renfro and Sargent (2017) for basket/umbrella trials; all N-of-1 frameworks.
- Wang and Schork (2019) and Schork (2022) also use Architecture A despite sharing authorship with Hendrickson et al. (2020), suggesting the MVN parameterization is a paper-specific choice, not a lab convention.
- To the best of our knowledge, no other clinical trial simulation framework uses differential correlation (MVN) to encode the biomarker-treatment interaction.

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The dominant approach in the clinical trial simulation literature is direct mean moderation, in which the biomarker enters the outcome model as an interaction term that explicitly scales the treatment effect. Simon [@simon2010] employs this approach throughout the simulation frameworks he develops for biomarker-stratified designs; Freidlin and Korn [@freidlin2014] adopt it for adaptive enrichment trials; Renfro and Sargent [@renfro2017] use it throughout their treatment of basket and umbrella trials; and it underlies the platform trial literature without exception. Recent simulation work on biomarker-by-treatment interaction estimation in randomized trials with prognostic covariates, specifically the study by Haller and Ulm [@haller2018], likewise adopts the standard regression specification with explicit interaction terms. Among N-of-1 simulation studies, Zucker et al. [@zucker1997], Araujo et al. [@araujo2016], and Duan et al. (2013) similarly use hierarchical random-effects models in which the biomarker predicts the individual treatment effect through the mean structure. Notably, Wang and Schork [@wang2019] and Schork [@schork2022], whose methodological work addresses power and design for N-of-1 trials with serial correlation and carryover directly, also specify the treatment effect through a linear mean structure, which suggests that the multivariate-normal differential-correlation parameterization of Hendrickson et al. [@hendrickson2020] is not a convention of any particular research program but rather a paper-specific design choice. To the best of our knowledge, no other clinical trial simulation framework generates the biomarker-treatment interaction through differential correlation in a multivariate normal joint distribution rather than through direct mean moderation.

[4] **The two architectures were discovered by accident during pmsimstats development, when two implementations of the same simulation diverged under carryover.**

- The original code uses Architecture B (MVN); a tidyverse reimplementation inadvertently used Architecture A.
- Both produced plausible power estimates, but they reached qualitatively different conclusions about carryover effects.
- Tracing the discrepancy to its root revealed the architectural distinction, which has not been explicitly discussed in the simulation methodology literature.

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We encountered both approaches in the course of developing the pmsimstats simulation framework for N-of-1 clinical trials [@hendrickson2020]. The original publication code uses the MVN differential correlation approach, an unusual choice that, as we shall demonstrate here, has substantive consequences for power estimation under carryover. A comprehensive audit of the codebase subsequently revealed that a parallel tidyverse reimplementation of the same simulation had inadvertently adopted the standard direct mean moderation approach. Both implementations produced plausible power estimates, yet they yielded qualitatively different conclusions about how carryover effects influence statistical power. Tracing this discrepancy to its root cause revealed the architectural difference described in this paper, which has not, to the best of our knowledge, been discussed explicitly in the simulation methodology literature.

[5] **This paper is the first systematic comparison of the two DGP architectures; we formalize, simulate, and give guidance.**

- We formalize the distinction between Architecture A (mean moderation) and Architecture B (differential correlation).
- We demonstrate power consequences through Monte Carlo simulation.
- We offer guidance for investigators on how to choose the appropriate architecture.

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We present here what we believe to be the first systematic comparison of these two DGP architectures for biomarker-treatment interaction power estimation under carryover. We shall formalize the distinction between them, demonstrate its consequences through Monte Carlo simulation, and offer guidance for investigators designing simulation studies for predictive biomarker-moderated treatment effects.

Related documents in this series. The mathematical derivation, code-level implementation details, four implementation attempts (three failed and one correct), and the empirical evaluation of the original Hendrickson[1] parameter grid under Architecture~A are documented in `19-mean-moderation-implementation-notes.tex` (with a brief markdown summary in `19-mean-moderation-implementation-notes.md`). A condensed empirical-results summary suitable for collaborator presentation is provided in `21-architecture-comparison-summary.md`.

3 2. Two DGP Architectures

Notation. Throughout this paper, B_i denotes the biomarker value of participant i, D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it} the biological response component of the outcome. These three symbols describe the data-generating-process layer. The analysis-model layer uses a related but distinct continuous drug exposure variable, `Dbc`, defined in Section 3.1. Code-style identifiers in typewriter font (`bm`, `Dbc`, `BR`) refer to specific variables in the pmsimstats codebase or to R formula syntax; mathematical symbols are used otherwise.

3.1 2.1 Architecture A: Direct Mean Moderation

In this approach the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is:

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it}$$

where D_{it} is the treatment indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit in the mean: participants with higher biomarker values have proportionally larger treatment effects.

Key properties:

- The interaction is **deterministic** given the biomarker value and treatment status.
- The biomarker moderation β_{bm} is a **population-level parameter** that applies uniformly to all participants.
- The signal for the interaction is in the **first moment** (conditional mean) of the outcome distribution.
- Adding noise, random effects, or carryover to D_{it} scales the entire treatment effect (including its biomarker-dependent component) proportionally. The **ratio** of drug effect between high-biomarker and low-biomarker participants is preserved.

  [6] **The multiplicative and additive forms of Architecture A are interchangeable for power calculations.**

  - The multiplicative form maps directly to the standard additive regression, the effect-modeling equation of Kent et al. (2020).
  - $\beta_3 = \beta_1 \cdot \beta_{bm}$; the additive form additionally allows a prognostic main effect.

- Both forms fall within Architecture A and are interchangeable for power calculations on the interaction coefficient.

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We note that the multiplicative form above maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature, specifically the effect modeling equation of Kent et al. [@kent2020path]:

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$$

The interaction coefficient β_3 in the additive form corresponds to the product $\beta_1 \cdot \beta_{bm}$ in the multiplicative form. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For the purpose of power calculations on the interaction coefficient, the two forms are interchangeable up to this prognostic-effect distinction, and both fall within Architecture A.

This architecture is used in the pedagogical simulation (`implementations/simple/simulation.R`) and in the exploratory carryover analyses (`simulation_carryover_spectrum.R`) of the pmsimstats project.

3.2 2.2 Architecture B: Differential Correlation (MVN)

In this approach the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with treatment-state-dependent correlation:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right)$$

where the covariance matrix Σ has the following treatment-state-dependent BM-BR correlation:

$$\text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & \text{if } D_{it} = 1 \text{ (on drug)} \\ c_{bm} \cdot e^{-\lambda \cdot t_{sd}} & \text{if } D_{it} = 0 \text{ (off drug)} \end{cases}$$

where t_{sd} is time since drug discontinuation. The exponential-decay form is the default; the limiting cases $\lambda = 0$ (no decay; correlation persists indefinitely during the off-drug phase) and $\lambda \to \infty$ (immediate decay; correlation drops to zero the instant the drug is discontinued) recover the two no-carryover model variants.

[7] **In Architecture B the interaction is in the covariance structure, not the mean; it emerges from the conditional distribution given the biomarker value.**

- The population mean of BR_{it} is the same for all participants; no mean-structure interaction exists.
- Given biomarker value b, the conditional expectation of BR scales with c_{bm}, creating an interaction-like structure.
- Correlation is c_{bm} on drug and decays as $c_{bm}e^{-\lambda t_{sd}}$ off drug; moderation attenuates during washout.

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The interaction is not in the mean structure: the population mean of BR_{it} is the same for all participants with the same treatment history. Instead it emerges from the **conditional distribution**: given a participant's biomarker value $B_i = b$, the conditional expectation of their biological response is:

$$E[BR_{it}|B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

This conditional expectation has an interaction-like structure: the biomarker value b modulates the expected BR. The effective correlation is c_{bm} on drug and $c_{bm} \cdot e^{-\lambda t_{sd}}$ during the washout; the moderation therefore persists with attenuated strength during off-drug periods and vanishes only in the limit of complete decay.

Key properties:

- The interaction is **probabilistic**: it manifests as a tendency, not a deterministic scaling.
- The biomarker moderation c_{bm} is a **correlation parameter** that governs the strength of the association.
- The signal for the interaction is in the **second moment** (covariance structure) of the joint distribution.
- Carryover effects in the DGP inflate the BR means during off-drug periods, making them resemble on-drug means. This reduces the **differential** in the correlation structure between treatment states, weakening the signal that the analysis model detects.

This architecture is used in the Hendrickson et al.[1] publication code and the audited `R/` package in pmsimstats-ng.

3.3 2.3 Architecture C: Combined Mean Moderation and Differential Correlation

[8] **Architecture C activates both interaction channels simultaneously, with independently calibrated weights.**

- The MVN covariance encodes BM-BR differential correlation with weight `c.bm_b` (the Architecture B channel).
- After the MVN draw, BR is additively shifted on-drug by $c_{bm,A} \cdot z_B \cdot \sigma_{BR}$ (the Architecture A channel).
- Special cases: $c_{bm,B} = 0$ recovers pure Architecture A; $c_{comb,A} = 0$ recovers pure Architecture B.

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Architecture C (combined) applies both interaction mechanisms in sequence. First, the MVN covariance matrix is constructed with BM-BR differential correlation weighted by `c.bm_b`, exactly as under Architecture B. Second, after the MVN draw, the BR component is additively shifted at on-drug timepoints by

$$\Delta BR_{it} = c_{bm,A} \cdot z_{B,i} \cdot \sigma_{BR} \cdot \mathbf{1}[\text{on-drug}]$$

where $z_{B,i} = (B_i - \mu_B)/\sigma_B$ is the standardized biomarker for participant i, exactly as under Architecture A.

The two parameters `c.bm_a` and `c.bm_b` are independent. Setting $c_{bm,B} = 0$ removes the covariance channel and recovers pure Architecture A; setting $c_{bm,A} = 0$ removes the mean-shift channel and recovers pure Architecture B. The boundary cases are enforced algebraically rather than by convention, so the special cases serve as internal validation gates for any simulation study using Architecture C.

We note that when both parameters are set to their full values ($c_{bm,A} = c_{bm,B} = 0.45$), the total interaction effect on the BR mean is approximately double that of either pure architecture at $c_{bm} = 0.45$. A researcher wishing to hold total effect size constant across architectures may instead parameterize as $c_{bm,A} = (1-\alpha) \cdot c$ and $c_{bm,B} = \alpha \cdot c$ for a mixing weight $\alpha \in [0, 1]$ and total weight c. We do not adopt this constraint in the simulations below, preferring the factorial parameterization that spans the full $(c_{bm,A}, c_{bm,B})$ space.

[9] **The biological rationale for Architecture C is that the two channels are mechanistically distinct and may both operate.**

- The mean-shift channel (A) reflects pharmacodynamic scaling: the drug's biological effect is larger in high-biomarker participants.
- The covariance channel (B) reflects differential co-regulation: on-drug periods couple biomarker expression to treatment response in a way that washes out between periods.
- Neither channel implies the other; a drug could activate one, both, or neither.

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The two channels are mechanistically distinct and need not scale in proportion. The mean-shift channel (Architecture A) reflects a pharmacodynamic relationship: the drug's biological effect on BR is amplified in participants whose biomarker is above the population mean, because the biomarker governs the effective dose or receptor occupancy. This mechanism operates at the level of the mean trajectory for each participant.

The covariance channel (Architecture B) reflects a different process: on-drug periods impose a co-regulatory coupling between biomarker expression and treatment response, perhaps because the drug activates a pathway whose output is gated by the biomarker signal. This coupling is absent during off-drug washout, producing the differential correlation structure that Architecture B encodes in the MVN covariance matrix.

A given drug could activate one, both, or neither channel. The simulation practitioner who is uncertain which channel predominates should regard Architecture C with a range of α values as the natural sensitivity framework.

3.4 2.4 Mathematical Comparison

[10] **Under Architecture A, carryover shrinks the signal amplitude but preserves the ratio between biomarker groups, so identifiability is maintained.**

- The expected outcome difference between two participants scales linearly with D.
- Under carryover, D is replaced by a decayed value and the absolute magnitude shrinks.
- But the ratio of drug effect between high- and low-biomarker participants is preserved at every exposure level; the signal contracts

but remains identifiable.

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We begin the comparison by considering the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D under Architecture A:

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D$$

The absolute magnitude of this difference scales linearly with D and therefore shrinks during off-drug periods when D is replaced by an exposure-decayed value $D \cdot \text{carryover}$. We note, however, that the **ratio** of the drug effect between high-biomarker and low-biomarker participants is preserved at every level of exposure. The biomarker-by-treatment signal contracts in amplitude but does not lose its identifiability under carryover.

[11] **Under Architecture B, the biomarker-dependent component of the off-drug response decays exponentially, eventually eliminating the differential signal.**

- Off-drug, the conditional expectation includes a biomarker-dependent term scaled by $c_{bm}e^{-\lambda t_{sd}}$.
- As t_{sd} grows, that term shrinks toward zero.
- In the limit, all participants converge to the same off-drug mean regardless of biomarker value; the interaction signal vanishes.

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Under Architecture B, the analogous quantity is:

$$E[BR|b_1, D = 1] - E[BR|b_2, D = 1] = c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b_1 - b_2)$$

During off-drug periods with carryover:

$$E[BR|b, D = 0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

Here the biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, eliminating the differential signal entirely.

[12] **Under Architecture C, carryover operates on both channels simultaneously; the power loss is bounded below by Architecture B and above by Architecture A.**

- The mean-shift channel degrades as under Architecture A: amplitude shrinks but identifiability is preserved at every exposure level.
- The covariance channel degrades as under Architecture B: off-drug BM-BR correlation decays toward zero, eroding second-moment signal.
- The net power loss lies between the two pure-architecture extremes, weighted by the relative magnitudes of $c_{bm,A}$ and $c_{bm,B}$.

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Under Architecture C both channels are active. The analysis model's interaction coefficient depends on the covariance between B_i and D_{bc}-weighted outcomes. The mean-shift channel contributes a term that degrades exactly as under Architecture A: the `Dbc` contrast is compressed during carryover but the ratio of the drug effect between biomarker groups is preserved at every exposure level. The covariance channel contributes a term that degrades as under Architecture B: the BM-BR correlation generating the second-moment signal decays exponentially off-drug. The two channels are additive at the level of the total biomarker-by-drug covariance, so the net power loss under Architecture C lies strictly between the pure-architecture extremes when $c_{bm,A}, c_{bm,B} > 0$. As either parameter approaches zero the corresponding boundary case is recovered.

---

4 3. Consequences for Statistical Power Under Carryover

4.1 3.1 Empirical demonstration

[13] **The two DGPs are identical except for how they encode the BM-BR linkage; any power divergence is attributable solely to that difference.**

- All non-biologic components (time-varying, placebo-belief, residual noise) are generated identically in both architectures.
- The only difference: Architecture B encodes BM-BR interaction in Σ; Architecture A adds a post-draw mean shift to on-drug BR.
- The same `nlme::lme` analysis model with `corCAR1` is used for both, so divergence in results is attributable to the DGP alone.

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We ran identical simulation configurations under both architectures using the pmsimstats-ng framework, matching the DGP parameters as closely as possible between the two approaches. We note that the non-biologic response components (time-varying response, placebo-belief accumulation, and residual noise) are generated from the same multivariate normal draw with identical means, variances, within-factor AR(1) autocorrelation, and cross-factor correlations in both architectures. The two DGPs differ *only* in how the biomarker-to-biological-response (BM-BR) linkage is encoded: Architecture B embeds it in the off-diagonal c_{bm} entries of Σ, whereas Architecture A omits those entries and adds a post-draw shift $c_{bm} \cdot B_i^* \cdot \sigma_{BR}$ to on-drug BR columns (see `19-mean-moderation-implementation-notes.tex` for the scaling derivation). The divergent carryover behavior reported below is therefore attributable to the BM-BR parameterization, not to any difference in how the placebo, time-varying, or residual-noise components are generated. The analysis model was the same in both cases: an `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation structure and a single biomarker-by-drug interaction term, written `bm:Dbc` in R formula syntax.

[14] **`Dbc` is a continuous drug exposure variable that captures exponential washout; the `bm:Dbc` interaction tests whether biomarker moderates that exposure-decayed effect.**

- `Dbc` $= 1$ on drug; $= \exp(-\lambda t_{sd})$ off drug, matching the DGP decay rate.
- Unlike the binary D_{it}, `Dbc` smoothly captures washout shrinkage.
- The test of `bm:Dbc` is the test of whether the biomarker moderates the (decay-adjusted) drug effect.

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The analysis-model drug exposure variable `Dbc` is the *continuous* analogue of the binary DGP indicator D_{it}. It equals 1 during on-drug periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time since drug discontinuation and λ is the analysis-model carryover decay rate (typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2 / t_{1/2}$, in which case it matches the DGP decay rate of Section 2.2 by construction). Whereas D_{it} flips abruptly between 0 and 1, `Dbc` captures the smooth shrinkage of residual drug effect during washout. The interaction coefficient on `bm:Dbc` is therefore the test of whether the biomarker moderates the exposure-decayed drug effect.

We follow the ADEMP framework of Morris, White, and Crowther (2019): Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.

Setup:

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation)
- Sample size: $N = 35$
- Biomarker correlation / moderation: $c_{bm} = 0.45$ / $\beta_{bm} = 0.45$
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks
- Analysis: matched `Dbc` with same half-life
- Replicates: 1,000 per cell (Monte Carlo standard error on a binomial power estimate near 0.5 is approximately $\sqrt{0.5 \cdot 0.5/1{,}000} \approx 0.016$)
- Implementation: 8-core `parallel::mclapply`; full factorial 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels = 36 cells, 36,000 fits, 100% convergence

[15] **$N = 35$ was chosen because it keeps power off the ceiling, making any carryover-induced loss visible; type I error is in the expected range.**

- $N = 35$ is the smallest prazosin-PTSD-calibrated sample size in the literature.
- Neither architecture saturates power at zero carryover, so losses are mechanically observable.
- Type I error across 18 null cells sat in $[0.010, 0.074]$, consistent with the conservative-to-nominal range expected for this specification.

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The $N = 35$ choice is the smallest of the prazosin-PTSD- calibrated sample sizes used in the published methodological literature; it is a regime in which neither architecture saturates power at the no-carryover cells, so any carryover-induced erosion is mechanically visible. Type I error under the $c_{bm} = 0$ null cells sat in $[0.010, 0.074]$ across the 18 null cells, consistent with the conservative-to- nominal regime expected for the continuous D_{bc} specification at this sample size.

Results under Architecture A (direct mean moderation):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.992	0.987	0.978
OL+BDC	0.751	0.748	0.730

[16] **Architecture A shows modest power loss under carryover (1–3%), well below the broader literature's 8–13% range.**

- Relative losses: 1.9% (CO), 1.4% (Hybrid), 2.8% (OL+BDC).
- All three fall below the 8–13% range reported in the broader N-of-1 simulation literature.
- Qualitative finding: modest erosion; power essentially intact under carryover.

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Power declines modestly under Architecture A: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 1.9% (CO), 1.4% (Hybrid), and 2.8% (OL+BDC). We note that all three values fall well below the 8-13% relative-loss range that the broader N-of-1 simulation literature reports for direct-mean-moderation DGPs, though they reproduce its qualitative finding, namely a modest erosion that leaves power essentially intact.

Results under Architecture B (differential correlation, MVN):

[17] **Architecture B shows sharply larger power loss under carryover, especially OL+BDC (30.6%); the B-vs-A gap is significant for OL+BDC and Hybrid but not CO.**

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.995	0.980	0.944
OL+BDC	0.777	0.694	0.539

- Relative losses: 3.0% (CO), 5.1% (Hybrid), 30.6% (OL+BDC); the OL+BDC loss is $\approx 11\times$ the matched Architecture A loss.
- B-vs-A gap: $z = 7.64$ ($p < 10^{-13}$) at OL+BDC; $z = 3.96$ ($p < 10^{-4}$) at Hybrid.
- CO gap not significant ($z = 0.29$): the CO design's within-subject AR(1) structure dominates the correlation-decay channel that Architecture B depends on.

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Power declines more sharply under Architecture B, and most markedly so in the OL+BDC design: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 3.0% (CO; not significantly different from Architecture A's CO loss), 5.1% (Hybrid), and **30.6% (OL+BDC)**. The CO cell shows a small, non-monotone fluctuation ($0.745 \to 0.754 \to 0.723$), which is within one Monte Carlo standard error and is consistent with the null finding of no significant architecture-by-carryover interaction in that design. The OL+BDC loss is approximately 11 times the matched Architecture-A OL+BDC loss (2.8%). It falls below the 40-60% relative-loss range that the broader N-of-1 simulation literature reports for the MVN differential-correlation DGP, but it confirms the qualitative finding, namely a substantial and strongly design-dependent erosion. We compare the carryover-induced power loss between architectures using a difference-of-differences z-statistic. Let $\Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}$ denote the within-architecture loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ for architecture $X \in \{A, B\}$, estimated from $n = 1{,}000$ independent replicates at each cell. The B-versus-A gap is $\Delta_B - \Delta_A$, with standard error

$$\mathrm{SE} = \sqrt{\frac{\hat{\pi}_{B,0}(1-\hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1-\hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1-\hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1-\hat{\pi}_{A,1})}{n}}$$

since the four proportions come from independent replicate sets. All tests are two-sided. At OL+BDC this gives $z = 7.64$ ($p < 10^{-13}$); at Hybrid, $z = 3.96$ ($p < 10^{-4}$); the CO contrast is not statistically distinguishable from zero ($z = 0.29$) because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation decay channel that Architecture B uses to encode the biomarker-treatment interaction.

4.2 3.2 Why the architectures diverge

We turn now to a mechanistic account of why the two architectures produce such different power losses under carryover.

[18] **Architecture A is subject to only one carryover channel (mean blurring); it escapes the second channel (correlation erosion).**

- Carryover compresses `Dbc` contrast and inflates off-drug BR means; both architectures share this channel.
- Architecture A lacks a second channel because β_{bm} is a fixed population parameter and the noise structure is drug-state-independent.
- Result: reduced variance in the interaction term, but signal-to-noise ratio degrades only modestly.

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Under Architecture A, carryover reduces the magnitude of the drug exposure variable seen by the analysis model (`Dbc` drops below 1 during washout), and the mean BR at off-drug timepoints is inflated by the residual drug effect exactly as in Architecture B. Both mechanisms compress the between-state treatment contrast and are shared by the two architectures. What Architecture A lacks is the second channel of signal loss: the biomarker moderation coefficient β_{bm} is a fixed population parameter, the biomarker enters the mean structure directly rather than through a correlation, and the noise structure is independent of drug state. The biomarker-by-drug interaction term therefore has reduced variance (because `Dbc` has less contrast between drug states), but the signal-to-noise ratio degrades only modestly.

Under Architecture B, carryover operates on two levels simultaneously:

1. **Mean blurring**: The BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means. This reduces the treatment contrast that the analysis model uses to distinguish drug effect from no-drug. This channel is shared with Architecture A.
2. **Correlation erosion**: The BM-BR correlation during off-drug periods de-

cays exponentially with t_{sd}. This is not just a statistical modeling choice; it is a structural consequence of the DGP. The correlation between biomarker and BR reflects the degree to which the drug effect is active; as the drug washes out, the correlation weakens because the drug-mediated component of BR variance shrinks relative to the drug-independent component. This channel is *unique* to Architecture B, because only here does the biomarker signal live in the second moment of the joint distribution. Under Architecture A the drug-mediated and drug-independent components of BR variance move in tandem (both are scaled by the same `Dbc`-weighted drug effect), so their ratio, and hence the identifiability of the interaction, is preserved even as amplitudes shrink.

[19] **Under Architecture B the interaction estimate is attacked from both sides simultaneously; Architecture A faces only one of the two attacks.**

- The `bm:Dbc` coefficient depends on the covariance between B_i and `Dbc`-weighted outcomes.
- Under Architecture B: (1) `Dbc` contrast is compressed; (2) the BM-BR correlation generating the covariance signal is simultaneously decaying.
- Architecture A faces only attack (1), not attack (2); this single difference accounts for the divergence in power.

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The analysis model's interaction coefficient (the coefficient on `bm:Dbc`) depends on the covariance between B_i and `Dbc`-weighted outcomes. Under Architecture B, this covariance is being attacked from both sides: the treatment contrast in `Dbc` is compressed, and the BM-BR correlation that generates the covariance signal is simultaneously decaying. That Architecture A escapes the second attack entirely accounts for the divergence in power loss documented above.

4.3 3.3 Robustness check at $N = 70$

[20] **An $N = 70$ confirmation run checks whether the qualitative ordering holds when power is near the ceiling.**

- 800 replicates per cell, 12 cells (2 architectures $\times$ 2 c_{bm} $\times$ 3 $t_{1/2}$), OL+BDC design.
- 100% convergence across 9,600 fits.
- Purpose: verify the Section 3.1 ordering survives saturation at the larger N.

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The Section 3.1 production results are at $N = 35$, where neither architecture saturates power and the carryover-induced loss is mechanically visible. We conducted a confirmation run at $N = 70$ to check whether the qualitative ordering survives the saturation that comes with the larger sample size. An 800-replicate-per-cell Monte Carlo run was executed at $N = 70$ across 12 cells (architecture $\in \{\text{MVN}, \text{mean moderation}\} \times c_{bm} \in \{0, 0.45\} \times t_{1/2} \in \{0, 0.5, 1.0\}$ weeks), all on the OL+BDC design, using 8-core `parallel::mclapply`. Wall-clock 483 s; convergence 100% across 9,600 fits.

At $c_{bm} = 0.45$:

Architecture	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
MVN (B)	0.978	0.945	0.885
Mean moderation (A)	0.976	0.973	0.959

[21] **The $N = 70$ run confirms the qualitative ordering; compressed absolute magnitudes reflect ceiling saturation, not a smaller underlying effect.**

- Losses: 9.3 pp (Architecture B, $p < 10^{-12}$) vs. 1.7 pp (Architecture A, $p \approx 0.05$); ordering matches $N = 35$.
- At $N = 70$ both architectures are within 2–5 pp of the ceiling at $t_{1/2} = 0$, mechanically truncating observable loss.
- $N = 35$ production results remain the canonical reference for the narrative magnitude claims.

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Within-architecture losses from $t_{1/2} = 0$ to $t_{1/2} = 1$ are 9.3 percentage points (Architecture B; two-proportion z-test on the paired cells, $z = 7.4$, $p < 10^{-12}$) and 1.7 percentage points (Architecture A; $z = 2.0$, $p \approx 0.05$), where each z-statistic uses the formula $(\hat{\pi}_0 - \hat{\pi}_1)/\sqrt{\hat{\pi}_0(1-\hat{\pi}_0)/n + \hat{\pi}_1(1-\hat{\pi}_1)/n}$ with $n = 800$ replicates per cell. The qualitative ordering matches Section 3.1's $N = 35$ result. The absolute magnitudes are compressed because $N = 70$ at the prazosin-PTSD calibration places both architectures within

2-5 percentage points of the unit ceiling at $t_{1/2} = 0$, mechanically truncating the loss either architecture's curve can absorb. We note that reading the narrative loss-magnitude claim against the $N = 70$ robustness numbers (3% and 9% relative loss for Architecture B versus 0.4% and 1.8% for Architecture A) requires acknowledging this saturation effect; the $N = 35$ production results in Section 3.1 are the canonical reference for the narrative magnitude.

[22] **Type I error is well-controlled across all 18 null cells, with no architecture-specific inflation.**

- Type I error range $[0.010, 0.074]$, mean 0.043, median 0.042 across 18 null cells at $N = 35$.
- No architecture-specific inflation detected.
- The slightly conservative Architecture B OL+BDC null cell at $t_{1/2} = 1.0$ ($p = 0.010$) is the only notable departure.

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Type I error. Across all 18 null cells of the $N = 35$ production run, type I error sat in $[0.010, 0.074]$, with mean 0.043 and median 0.042. We observe no architecture-specific inflation. Architecture B's $t_{1/2} = 1.0$ OL+BDC null ($p = 0.010$, slightly conservative) is the smallest cell; the corresponding alternative ($p = 0.539$) is the most informative cell of the Section 3.1 narrative.

[23] **Estimator bias is small and architecturally interpretable; coefficient drift under Architecture A contrasts with correlation-channel erosion under Architecture B.**

- Mean $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranges -0.231 to -0.281 across alternative cells; Architecture A OL+BDC at $t_{1/2} = 1.0$ produces the largest absolute mean.
- Architecture B cells maintain mean near -0.234 across all $t_{1/2}$ values, consistent with erosion through correlation decay rather than coefficient drift.
- Nominal-95% Wald coverage holds across the grid.

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Estimator bias. Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from

-0.231 to -0.281 across the 18 alternative cells, with the $N = 35$ Architecture-A OL+BDC cell at $t_{1/2} = 1.0$ producing the largest absolute mean (-0.281); the Architecture B cells maintained mean near -0.234 across all $t_{1/2}$ values, consistent with the Architecture-B information channel being eroded through correlation decay rather than through coefficient drift. Empirical SE of $\widehat{\beta}_{bm:D_{bc}}$ runs 0.05 to 0.10 across cells; the within-replicate model SE matches at scale, indicating nominal-95% Wald coverage holds across the grid.

4.4 3.4 Architecture C: combined-channel power results

[24] **Architecture C power lies between the pure-architecture extremes; the 3×3 parameter panel quantifies how the two channels trade off.**

- Grid: $c_{bm,A} \times c_{bm,B} \in \{0, 0.22, 0.45\}^2$; boundary rows/columns correspond to pure Architecture A and B respectively.
- As $c_{bm,B}$ increases from 0 to 0.45, power loss under carryover intensifies monotonically.
- As $c_{bm,A}$ increases from 0 to 0.45, the mean-shift channel adds signal that is resilient to carryover, partially offsetting the covariance-channel loss.

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We ran a factorial simulation on the Architecture C parameter grid. The design factors are CO, Hybrid, and OL+BDC; $N = 35$ total subjects per cell (randomised equally across paths); $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks; 1000 replicates per cell; A2 analysis specification. The parameter panel is $c_{bm,A} \times c_{bm,B} \in \{0, 0.22, 0.45\}^2$ (nine cells per design–carryover combination). By construction, the boundary cells ($c_{bm,A} = 0.45$, $c_{bm,B} = 0$) reduce the combined DGP to pure Architecture A, and ($c_{bm,A} = 0$, $c_{bm,B} = 0.45$) reduce it to pure Architecture B; these boundary columns and rows therefore serve as internal consistency gates confirming that the combined parameterisation recovers each limiting case algebraically.

[25] **The 3×3 panel confirms the mathematical prediction: Architecture C power is a monotone function of both parameters, with interior cells super-additive on the probability scale.**

- Interior cells ($c_{bm,A} > 0$, $c_{bm,B} > 0$) exceed the probability-scale additive benchmark $p(c_A, 0) + p(0, c_B) - p(0, 0)$.

Table 1: Architecture C empirical power: 3×3 parameter panel. $N = 35$, carryover $t_{1/2} = 1.0$ week, A2 analysis specification, 1000 replicates per cell. Rows index $c_{bm,A}$ (mean-shift channel); columns index $c_{bm,B}$ (covariance channel). By construction, the boundary column $c_{bm,B} = 0$ corresponds to pure Architecture A; the boundary row $c_{bm,A} = 0$ corresponds to pure Architecture B.

	$c_{bm,B}$ (covariance channel)		
$c_{bm,A}$ (mean channel)	0.00	0.22	0.45
Crossover (CO)			
0.00	0.046	0.064	0.206
0.22	0.070	0.179	0.361
0.45	0.172	0.354	0.528
Hybrid			
0.00	0.034	0.083	0.273
0.22	0.108	0.299	0.621
0.45	0.389	0.642	0.870
OL+BDC			
0.00	0.017	0.049	0.148
0.22	0.086	0.164	0.362
0.45	0.276	0.473	0.688

- Moving along the $c_{bm,B}$ axis (increasing covariance weight, fixed mean-shift weight) degrades power under carryover; moving along the $c_{bm,A}$ axis attenuates that degradation.
- The equal-weight cell ($c_{bm,A} = c_{bm,B} = 0.45$) has higher baseline power than either pure architecture at $c_{bm} = 0.45$, reflecting the doubled total signal, but also higher absolute carryover sensitivity than Architecture A alone.

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Table 1 reports empirical power for Architecture C across the full 3×3 parameter panel at the most demanding carryover condition ($t_{1/2} = 1.0$ week). Several features of the results merit comment.

First, the monotone structure is exact: power increases strictly with each channel parameter, holding the other fixed, across all three designs. This confirms that the two channels contribute independently and additively to detectability.

Second, interior cells exceed the probability-scale additive benchmark. At ($c_{bm,A} = 0.22$, $c_{bm,B} = 0.22$) in the Hybrid design, power is 0.299. The additive baseline on the probability scale – the power expected if the two channels contributed independently – is $p(0.22, 0)+p(0, 0.22)-$

$p(0,0) = 0.108 + 0.083 - 0.034 = 0.157$. The observed value of 0.299 exceeds this benchmark by 0.142, indicating genuine super-additivity: the joint signal compounds rather than aggregates.

Third, the covariance channel ($c_{bm,B}$) and the mean-shift channel ($c_{bm,A}$) contribute comparably to carryover sensitivity. Comparing the no-carryover ($t_{1/2} = 0$) and one-week ($t_{1/2} = 1.0$) panels for the CO design, increasing $c_{bm,B}$ from 0 to 0.45 (fixing $c_{bm,A} = 0$) degrades power from 0.368 to 0.206 – a 44% relative loss. The corresponding loss along the $c_{bm,A}$ axis (fixing $c_{bm,B} = 0$) is from 0.352 to 0.172 – a 51% loss, in the same range. At the equal-weight interior cell ($c_{bm,A} = c_{bm,B} = 0.45$), the Hybrid design retains 0.870 power under one-week carryover, substantially exceeding either pure architecture alone at $c_{bm} = 0.45$ (Architecture A: 0.389; Architecture B: 0.273).

Fourth, design choice remains consequential under Architecture C. At the full-signal, one-week-carryover cell, Hybrid (0.870) substantially outperforms OL+BDC (0.688) and especially CO (0.528). The design ranking established in Section 3.2 is preserved throughout the Architecture C grid.

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5 4. Biological Assumptions

[26] **Architecture choice encodes a biological claim about how the biomarker moderates treatment response.**

- The three architectures reflect distinct assumptions about the mechanism of biomarker-treatment moderation.
- Section 4 takes each architecture in turn (4.1, 4.2, 4.3 for Architectures A, B, and C respectively), then discusses the prazosin-PTSD motivating case.

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The choice between architectures is not merely a statistical convenience; it reflects different assumptions about the biological mechanism by which the biomarker moderates treatment response. We consider each architecture in turn, then take up the prazosin-PTSD motivating application in detail.

5.1 4.1 When Architecture A is appropriate

[27] **Architecture A is appropriate when the biomarker directly determines the magnitude of the drug effect; the scaling is deterministic at the mean-structure level.**

- The drug effect scales with the biomarker value: $\text{effect}_i = f(B_i) \cdot \text{drug}$.
- 'Deterministic' means any two participants with the same B_i have the same expected response; individual outcomes still carry noise.
- Appropriate when the biomarker is a direct PK/PD determinant (eGFR, receptor density, metabolizer status).

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Architecture A (direct mean moderation) is appropriate when the biomarker governs the **magnitude of the drug's biological effect** for each individual participant. The drug effect for participant i is scaled by their biomarker value: $\text{effect}_i = f(B_i) \cdot \text{drug}$. The scaling is deterministic in the population-mean sense, in that any two participants sharing the same B_i have the same expected response, but individual outcomes still carry the usual noise, random-effect, and residual-biological variability. The label 'deterministic' refers to the mean structure of the DGP, not to the realized response of any single participant.

Clinical examples:

- **Renal clearance of a renally-excreted drug.** Estimated glomerular filtration rate (eGFR) directly governs the elimination rate of drugs such as digoxin or aminoglycoside antibiotics. Participants with higher eGFR clear the drug more rapidly, achieving lower steady-state concentrations and a proportionally smaller drug effect. The biomarker-drug relationship is mechanistic and deterministic: given the biomarker value, the effective dose at steady state is fully determined (up to random noise).

- **Receptor density moderation.** A biomarker that measures target receptor density (e.g., PET imaging of receptor availability for an antagonist) directly determines the pharmacodynamic response to that antagonist. More receptors means more drug effect, proportionally.

- **Genetic metabolizer status.** CYP2D6 metabolizer phenotype determines the rate of drug activation or clearance. Poor metabolizers of a prodrug receive less active compound, reducing their treatment effect proportionally.

- **Dose-response with biomarker-dependent effective dose.** When the biomarker determines the effective dose (through body composition, protein binding, or volume of distribution), the treatment effect scales with the biomarker through the dose-response curve.

[28] **In all Architecture A cases, the proportional biomarker-drug relationship is preserved during carryover washout.**

- Residual drug effect during washout maintains the same proportional relationship with the biomarker.
- If participant A has 2× the drug effect of B on-drug, they also have ≈2× the residual effect during washout.
- This preserved proportionality is what Architecture A encodes and why it loses little power under carryover.

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In all these cases the biomarker's moderating effect is preserved under carryover because the residual drug effect during off-drug periods maintains the same proportional relationship with the biomarker. If participant A has twice the drug effect of participant B when on drug, they will also have approximately twice the residual effect during the washout period. That proportionality is precisely what Architecture A encodes.

5.2 4.2 When Architecture B is appropriate

[29] **Architecture B is appropriate when the biomarker is a probabilistic predictor of response, mediated by latent factors the biomarker imperfectly indexes.**

- Appropriate when the biomarker predicts which participants are likely to respond, not the magnitude of any individual's response.
- The biomarker-response association is population-level and probabilistic, mediated by latent factors (disease subtype, neurobiological heterogeneity, comorbidities).
- Two participants with identical biomarker values may respond very differently.

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Architecture B (differential correlation) is appropriate when the biomarker predicts **which participants are likely to respond**, but the magnitude of any individual's response is not deterministically governed by the biomarker. The biomarker and the drug response are associated at the population level, but the association is mediated by latent factors (disease subtype, neurobiological heterogeneity, comorbidities) that the biomarker imperfectly indexes.

Clinical examples:

- **Blood pressure as a PTSD subtype marker.** Elevated resting blood pressure may index a noradrenergic-predominant PTSD phenotype [@hendrickson2020] that is more likely to respond to prazosin (an alpha-1 adrenergic antagonist). The blood pressure does not *cause* the drug to work better; rather, it correlates with an underlying neurobiological state that determines drug responsiveness. Two participants with identical blood pressure may have very different responses because blood pressure is an imperfect proxy for the noradrenergic state.
- **Baseline disease severity as a treatment predictor.** Trials in Alzheimer's disease, depression, and chronic pain often find that patients with more severe baseline symptoms show larger treatment effects (the 'floor effect' or 'regression to the mean' phenomenon). The baseline severity score correlates with treatment response, but the relationship is statistical (population-level tendency), not deterministic.
- **Inflammatory biomarkers in psychiatry.** C-reactive protein (CRP) or interleukin-6 levels may predict response to anti-inflammatory augmentation of antidepressants [@raison2013]. High-CRP patients are more likely to respond, but the relationship is probabilistic: many high-CRP patients do not respond, and some low-CRP patients do.
- **Genetic risk scores.** Polygenic risk scores predict disease trajectory and treatment response, but explain only a fraction of the variance. The remainder reflects unmeasured genetic, epigenetic, and environmental factors.

[30] **Under Architecture B, carryover weakens the biomarker-response correlation because the biomarker's predictive power depends on the presence of active drug.**

- As the drug washes out, the drug-mediated variance component diminishes, weakening the BM-BR correlation.
- The biomarker's predictive power is tied to active drug effect; without drug, it predicts nothing about non-drug symptom change.
- This is the 'correlation erosion' channel unique to Architecture B.

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In these cases carryover has a different implication: as the drug washes out, the correlation between biomarker and response weakens because the drug-mediated component of response variance diminishes. The biomarker's predictive power is inherently tied to the presence of active drug effect. Without drug, the biomarker is just a biomarker; it does not predict the non-drug component of symptom change.

[31] **Architecture B is a tractable second-moment approximation to the biologically faithful finite-mixture DGP; adequate for power calculations but not a mechanistic endorsement.**

- If the biomarker indexes a discrete responder class, the faithful DGP is a finite mixture; Architecture B approximates its population-level covariance structure.
- Architecture B captures the covariance implications of latent-class membership without committing to the full generative form.
- Adequate for power calculations; should not be read as endorsing MVN differential correlation as the correct mechanism.

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The latent-subtype framing raises a legitimate modeling question: if the biomarker is really a noisy indicator of an underlying responder class, the biologically faithful DGP would be an explicit finite mixture in which each participant is drawn from one of two (or more) response distributions with class-membership probability a function of B_i. Under such a mixture DGP the individual-level moderation is discrete, not probabilistic, and the MVN differential-correlation parameterization is best understood as a tractable second-moment approximation to a richer mixture model rather than as the generative mechanism itself. The practical consequence is that Architecture B as implemented here captures the population-level covariance implications of a latent-class mechanism without committing to its full generative form; this is adequate for power calculations that test for the presence of an interaction but should not be read as an endorsement of MVN differential correlation as the correct mechanistic model. Whether the observed power loss under carryover is better mitigated by analysis strategies (Section 6) or by explicit mixture modeling is an open question.

[32] **Senn (2016) cautions that apparent personalized response is often a variance artifact; Architecture B presupposes a real, stable latent subtype, which may not hold.**

- Senn (2016) argues that apparent biomarker-by-treatment interactions are often variance-component artifacts, not genuine effects.
- Architecture B presupposes the latent subtype is real, identifiable, and stable across the trial period.
- If that presupposition is shaky, the documented power loss may overstate recoverable signal.

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The latent-subtype framing of Architecture B should also be adopted with appropriate epistemic caution. Senn [@senn2016mastering] has argued that the apparent 'personal element in response to treatment' is often a variance-component artifact rather than a genuine biomarker-by-treatment interaction, and that even modest within-subject variability can produce convincing-looking but spurious differential response. Architecture B presupposes that the latent subtype indexed by the biomarker is real, identifiable, and stable across the trial period. When that presupposition is shaky, the apparent power loss under carryover documented in Section 3 may overstate what could be recovered under any analysis model, because the underlying signal it is attempting to estimate may itself be smaller than assumed.

5.3 4.3 The prazosin-PTSD case

We turn briefly to the motivating clinical application for pmsimstats, which is the active-duty prazosin trial by Raskind et al.[2], whose SBP-stratified secondary analysis is the subject of ongoing reanalysis by the pmsimstats team. The biomarker is resting systolic blood pressure (SBP) and the outcome is CAPS (Clinician-Administered PTSD Scale) score change.

[33] **The prazosin-PTSD evidence base is mixed; the PACT null result motivates the search for a predictive biomarker, with SBP as the leading candidate.**

- Earlier single-site Raskind trials (2003, 2007, 2013) reported benefit on nightmares and PTSD symptoms.
- PACT (2018) failed to replicate; this has been interpreted as consistent with effect heterogeneity.

- SBP, as a noninvasive proxy for central noradrenergic tone, is the leading predictive biomarker candidate.

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The clinical evidence base for prazosin in PTSD is mixed. The earlier single-site trials by Raskind and colleagues [@raskind2003; @raskind2007; @raskind2013] reported benefit on nightmares and global PTSD symptoms in veteran and active-duty samples. The subsequent multisite Veterans Affairs Cooperative Study Program trial [@raskind2018], known as PACT, did not replicate the effect: prazosin failed to outperform placebo on the primary endpoints. The negative PACT result has been interpreted by several investigators as consistent with effect heterogeneity, namely that prazosin works in some patients but not others, which motivates the search for a predictive biomarker that would identify the responder subgroup. Resting SBP, as a noninvasive proxy for central noradrenergic tone, is the leading candidate.

[34] **The SBP-prazosin mechanism is an Architecture B scenario: SBP is a proxy for noradrenergic state, not a direct PK determinant.**

- Elevated SBP indexes heightened noradrenergic tone, a PTSD subtype preferentially responsive to prazosin.
- SBP is a proxy, not a causal determinant; two patients with identical SBP may have very different noradrenergic states.
- This is precisely the Architecture B scenario: probabilistic association, not deterministic scaling.

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The biological hypothesis is that elevated SBP indexes heightened central noradrenergic tone, which characterizes a subtype of PTSD that is preferentially responsive to alpha-1 adrenergic blockade by prazosin. This is an Architecture B scenario: SBP is a proxy for an underlying neurobiological state, not a direct determinant of drug pharmacokinetics. Two patients with the same SBP may differ in their actual noradrenergic tone, alpha-1 receptor distribution, and downstream signaling, leading to different treatment responses despite identical biomarker values.

[35] **Hendrickson et al. (2020) correctly uses Architecture B for this context; the carryover power loss is clinically relevant for trial design decisions.**

- Architecture B (MVN differential correlation) is appropriate for the SBP-prazosin mechanism.
- The documented carryover sensitivity means designs with longer off-drug periods will have meaningfully less power.
- This effect is larger than a direct mean moderation simulation would predict.

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The Hendrickson et al. [@hendrickson2020] DGP uses Architecture B (MVN differential correlation) for this clinical context. The finding that carryover substantially reduces power under this architecture is therefore clinically relevant: it suggests that trial designs with longer off-drug periods will have meaningfully less power to detect the SBP-prazosin interaction than designs with shorter off-drug periods, and this effect is larger than what a direct mean moderation simulation would predict.

5.4 4.4 When Architecture C is appropriate

[36] **Architecture C is appropriate when both a pharmacodynamic-scaling pathway and a co-regulatory coupling pathway are plausible for the same biomarker.**

- The two channels can co-exist: prazosin's SBP-CAPS mechanism may involve both a mean-moderation pathway (SBP governs effective central noradrenergic suppression) and a differential-correlation pathway (on-drug periods couple noradrenergic tone more tightly to symptom variance).
- When the relative channel weights are unknown, the α-mixing parameterization of Architecture C is a natural sensitivity analysis spanning the range from pure A to pure B.
- Architecture C should not be the default; it has two free parameters where one suffices, and calibration requires independent evidence about both channels.

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Architecture C is appropriate when there is independent evidence that both the pharmacodynamic-scaling pathway and the co-regulatory coupling pathway are biologically active for the same biomarker-drug pair. Such evidence might include mechanistic assay data (e.g., the drug enhances biomarker-outcome correlation in vitro, consistent with the B channel, while receptor occupancy scales with biomarker level, consistent with the A channel) or replication across independent trials that are individually underpowered to distinguish the channels.

We note that Architecture C should not be the default choice for simulation studies, despite its generality. It carries two free parameters where either Architecture A or Architecture B requires only one, and calibrating both parameters independently demands more prior information than is typically available in the trial-design phase. The appropriate use cases are: (a) formal sensitivity analyses that ask how power estimates change across the $(c_{bm,A}, c_{bm,B})$ grid when the channel structure is genuinely uncertain; and (b) evaluation studies designed to determine which channel is dominant for a given biomarker-drug pair, using the simulation as a diagnostic instrument rather than a planning tool.

In the prazosin-PTSD context, Architecture C captures the possibility that SBP simultaneously governs the expected prazosin effect at the mean level (Architecture A, consistent with the pharmacological argument that noradrenergic tone determines the therapeutic window) and alters the degree of coupling between SBP and CAPS change during active drug exposure (Architecture B, consistent with a signaling-pathway interpretation). We regard Architecture C as the appropriate framework for evaluating the prazosin-SBP interaction if and when mechanistic data become available to calibrate both channels separately.

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6 5. Implications for Simulation Study Design

6.1 5.1 Choosing the appropriate architecture

The choice of DGP architecture should be guided by the biological mechanism hypothesized for the biomarker-treatment interaction:

[37] **When the channel structure is uncertain, Architecture C with a mixing-weight sensitivity sweep spans the range from**

Criterion	Architecture A	Architecture B	Architecture C
Biomarker role	Causal mediator	Statistical predictor	Both
Mechanism	Deterministic scaling	Probabilistic association	Both channels
Signal location	Mean structure	Covariance structure	Mean + covariance
Free parameters	1 (c_{bm})	1 (c_{bm})	2 ($c_{bm,A}$, $c_{bm,B}$)
Carryover impact	Modest (1–3%)	Design-dependent, up to 31%	Intermediate; scales with $c_{bm,B}$
Appropriate when	Biomarker determines dose or PK	Biomarker indexes a latent subtype	Channel uncertain; sensitivity use

pure A to pure B; Architecture B remains the conservative default.

- The α-mixing parameterization sets $c_{bm,A} = (1-\alpha)c$ and $c_{bm,B} = \alpha c$ for fixed total weight c and $\alpha \in [0, 1]$.
- $\alpha = 0$ recovers pure Architecture A; $\alpha = 1$ recovers pure Architecture B.
- Architecture B's larger carryover sensitivity makes it the safer default when the channel structure is unknown.

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If the biological mechanism is ambiguous, Architecture C with a range of mixing weights $\alpha \in [0, 1]$ (where $c_{bm,A} = (1 - \alpha)c$ and $c_{bm,B} = \alpha c$ for fixed total weight c) provides a natural sensitivity framework spanning the two pure-architecture extremes. Architecture B should remain the default conservative choice for trial design when the channel structure is unknown: its larger carryover sensitivity ensures that power estimates will not be systematically optimistic relative to either pure-architecture alternative.

6.2 5.2 Reporting requirements

Simulation studies evaluating biomarker-moderated treatment effects should explicitly state:

1. Whether the biomarker-treatment interaction is generated through mean moderation, differential correlation, or both channels simultaneously.
2. The biological rationale for the chosen mechanism and, for Architecture C, independent evidence supporting each channel.
3. Whether the carryover model (if present) operates on the interaction signal

consistently with the chosen architecture.

4. Sensitivity analyses under alternative architectures, if the biological mechanism is uncertain.

[38] **The four reporting requirements are the ADEMP 'D' element applied specifically to biomarker-validation simulations.**

- ADEMP (2019) prescribes five elements: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.
- The DGP-architecture choice concerns the D element; the four requirements above specialize that element for architecture-sensitive simulations.

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The reporting framework above is consistent with the broader ADEMP template of Morris, White, and Crowther [@morris2019simulation], which prescribes that simulation studies fix five elements before production runs: the **Aims** of the study, the **Data-generating mechanisms**, the **Estimands** (the inferential targets), the **Methods** (analysis specifications compared), and the **Performance measures** (power, bias, coverage, type I error, and Monte Carlo standard errors of each). The DGP-architecture choice this paper concerns is the **D** element of an ADEMP specification; the reporting requirements above are the specialisation of the broader ADEMP framework to the architecture-sensitive class of biomarker-validation simulations.

[39] **Primary estimand: $\beta_{bm:D}$ in the LME model; performance measures cover power, type I error, bias, SE, and convergence; 1000 replicates per cell.**

- Estimand: $\beta_{bm:D}$ in Sx $\sim bm + t + Dbc + bm\colon Dbc$ with random intercept and AR(1).
- Five performance measures: power, type I error, bias, empirical SE, convergence rate.
- $n_{\text{reps}} = 1000$ throughout; calibrated to achieve MCSE < 0.02 for power near 0.5.

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The **primary estimand** for this paper's simulations is the biomarker-by-treatment interaction coefficient $\beta_{bm:D}$ in the linear-mixed analysis model Sx $\sim bm + t + Dbc + bm : Dbc$ with random intercept and AR(1) residual correlation. The reported **performance measures** are: (i) power for $\beta_{bm:D}$ at $\alpha = 0.05$, with Monte Carlo standard error $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$; (ii) type I error under $c_{bm} = 0$, with the same MCSE; (iii) bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true value at each parameter cell; (iv) the empirical Monte Carlo standard error of $\hat{\beta}_{bm:D}$ across replicates; and (v) the convergence rate of the linear-mixed fit. Sample size, biomarker effect size, and carryover half-life vary across the parameter grid; the number of replicates per cell is calibrated to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. The simulations reported here use $n_{\text{reps}} = 1000$ throughout.

6.3 5.3 When the choice is uncertain

If the biological mechanism is ambiguous, as it often is in early biomarker development, we suggest investigators take the following steps:

1. Run the simulation under both architectures and report the range of power estimates.
2. Architecture B produces more conservative (lower) power estimates under carryover, making it the safer default for trial design decisions.
3. Design the trial to minimize carryover (adequate washout periods) to reduce sensitivity to the DGP architecture choice.

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7 6. Alternative Analysis Strategies Under Architecture B

This section provides a conceptual overview of six candidate strategies for mitigating the power loss documented in Section 3. None of the strategies below has been evaluated by simulation within this study; the assessments of expected benefit are qualitative and based on the mechanistic account in Section 3.2. A systematic simulation comparison is warranted before any strategy is recommended for applied use.

The power loss documented in Section 3 arises from two distinct sources: (1) compressed variance of the treatment indicator `Dbc` when carryover is present, and (2) erosion of the BM-BR correlation during off-drug periods. Source 1 is a property of the analysis model parameterization and can potentially be addressed by choosing a different predictor. Source 2 is a property of the data itself (the signal is genuinely weaker in off-drug observations), and no analysis model can recover signal that is not present. However, analysis strategies that are selective about *which observations contribute to the test* or *how they are weighted* can mitigate the damage.

7.1 6.1 Restrict analysis to on-drug observations

The most direct approach is to discard off-drug timepoints entirely and test the biomarker-treatment interaction using only on-drug observations. Every retained observation has the full BM-BR correlation (c_{bm}), so correlation erosion is eliminated. The costs are twofold: a reduction in effective sample size (fewer observations per participant) and, more importantly, the loss of the off-drug reference that separates the biological response from placebo and time-varying background response. Without off-drug observations the biological, placebo, and time-varying components are confounded, which is why the Open-Label design, despite having the most on-drug timepoints, does not dominate the other designs in the Section 3 results. For designs with many on-drug timepoints (OL has 8, the Hybrid design has 5-6 per path), the net effect may be positive: the per-observation signal-to-noise ratio is maximal, and the analysis model simplifies to `Sx ~ bm + t + bm:t` since there is no treatment contrast to model. The interaction is detected through the correlation between biomarker and the time-on-drug trajectory.

[40] **Restricting to on-drug observations eliminates correlation erosion but loses the off-drug reference; best for OL and Hybrid designs, worst for CO.**

- Benefit: every retained observation has full c_{bm}; correlation erosion is eliminated.
- Cost: loss of off-drug reference confounds biological, placebo, and time-varying components.
- Practical: best for OL/Hybrid (many on-drug timepoints); least useful for CO (discards half the observations).

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This approach is most promising for the Open-Label design (which has no off-drug observations regardless) and the Hybrid design (which has a long initial on-drug phase). It is least useful for the Crossover design, where discarding half the observations substantially reduces statistical power.

7.2 6.2 Weighted analysis

[41] **Weighted analysis downweights off-drug observations**

by drug exposure without discarding them; information-theoretic rationale; straightforward to implement.

- Weight: $w_t = e^{-\lambda t_{sd}}$ for off-drug observations; 1 for on-drug.
- Rationale: higher exposure = stronger correlation signal = more information about the interaction.
- Implemented via the `weights` argument in `nlme::lme`.

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Rather than discarding off-drug observations entirely, a weighted analysis retains all observations but weights them by their estimated drug exposure level. On-drug observations receive weight 1. Off-drug observations receive weight proportional to the expected residual drug effect:

$$w_t = e^{-\lambda \cdot t_{sd}}$$

This downweights observations where the BM-BR correlation has decayed (and where they contribute more noise than signal to the interaction estimate) without discarding them entirely. The rationale is information-theoretic: observations with higher drug exposure carry more information about the biomarker-treatment interaction because the correlation signal is stronger. This is straightforward to implement in `nlme::lme` using the `weights` argument.

7.3 6.3 Within-subject contrast

For designs with both on-drug and off-drug periods (CO, Hybrid, OL+BDC), a summary-measure approach computes a per-participant contrast: the difference between mean on-drug response and mean off-drug response. The biomarker-treatment interaction is then tested by a simple regression of these contrasts on the biomarker:

$$\bar{Y}_{i,\text{on}} - \bar{Y}_{i,\text{off}} = \alpha + \gamma \cdot B_i + \epsilon_i$$

[42] **Within-subject contrast sidesteps the repeated-measures model; trades efficiency for robustness to carryover misspecification.**

- Per-participant contrast $\bar{Y}_{\text{on}} - \bar{Y}_{\text{off}}$ is regressed on the biomarker.
- Carryover compresses the contrast magnitude but the contrast-biomarker correlation may be more robust.

- Trade-off: less efficient than a full longitudinal model, but more robust to carryover and correlation misspecification.

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This sidesteps the `Dbc` parameterization and the repeated-measures model entirely. Carryover affects the magnitude of the contrast (making off-drug means closer to on-drug means), but the *correlation between the contrast and the biomarker* may be more robust because within-subject averaging reduces noise before the biomarker association is tested. The approach trades the efficiency of a full longitudinal model for robustness to misspecification of the carryover and correlation structure.

7.4 6.4 Two-stage random slopes

A two-stage approach first estimates participant-specific treatment effects, then tests their association with the biomarker. Stage 1 fits a random-slopes model:

$$Y_{it} = \beta_0 + \beta_1 t + \beta_2 D_{it} + u_{0i} + u_{1i} D_{it} + \epsilon_{it}$$

where β_2 is the population mean drug effect and u_{1i} is participant i's random deviation from that mean. The participant-specific drug effect is then $\beta_2 + u_{1i}$. Stage 2 regresses the estimated random deviations (or equivalently, the BLUPs of the participant-specific effects) on the biomarker:

$$\hat{u}_{1i} = \delta_0 + \delta_1 B_i + \eta_i$$

[43] **Two-stage random slopes separates within-subject effect estimation from between-subject biomarker association; may be more robust to carryover.**

- Stage 1: random-slopes LME estimates participant-specific drug effects ($\beta_2 + u_{1i}$).
- Stage 2: regress BLUPs on biomarker to test association.
- The random slope u_{1i} integrates over the full trajectory; robustness to carryover is plausible but unverified by simulation.

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This separates the within-subject treatment effect estimation (which uses the longitudinal data and is affected by carryover) from the between-subject biomarker association (which uses the cross-sectional relationship between estimated effects and biomarker values). The approach may be more robust to carryover because the random slope u_{1i} integrates over the participant's entire trajectory, including both on-drug and off-drug phases.

7.5 6.5 Exclusion of contaminated observations

[44] **Excluding the first 1–2 off-drug timepoints removes the most contaminated observations; later off-drug timepoints provide a cleaner contrast.**

- Early off-drug timepoints: highest residual drug effect, BM-BR correlation actively decaying.
- Later off-drug timepoints: fully washed out, cleaner contrast with on-drug phase.
- Result: comparison between fully-on and fully-off states, avoiding the strained intermediate zone.

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The first 1-2 off-drug timepoints carry the most carryover contamination: the residual drug effect is highest, and the BM-BR correlation is in the process of decaying. Later off-drug timepoints, where the drug has fully washed out, provide a cleaner contrast with the on-drug phase. Excluding the early off-drug observations and retaining only the later ones yields a comparison between fully-on and fully-off states, avoiding the intermediate zone where the analysis model is most strained.

7.6 6.6 Design-level solutions

Rather than adapting the analysis model to accommodate carryover, the trial design itself can be modified to reduce carryover exposure:

- **Longer washout periods** between drug phases eliminate residual drug effect before off-drug measurements begin.
- **More on-drug timepoints** relative to off-drug increase the proportion of high-signal observations.
- **Front-loaded drug exposure** (as in the OL+BDC design) places a long initial on-drug block before the off-drug baseline-during-continued-

treatment phase, so that the BM-BR correlation has been fully established before any off-drug observations are collected. The alternative ordering (off-drug first, then on-drug) has the appealing property of placing the placebo and time-varying estimation in an uncontaminated window but sacrifices on-drug exposure duration and forces the BM-BR signal to be re-established after a wash-in; the net power trade-off depends on the ratio of carryover half-life to total trial length and is explored empirically in the Section 3 results.

[45] **Design-level changes address the root cause rather than the symptom; cost is longer trial or reduced ability to estimate carryover dynamics.**

- Three levers: longer washout, more on-drug timepoints, front-loaded drug exposure (as in OL+BDC).
- Front-loading establishes BM-BR correlation before any off-drug observations; alternative ordering (off-drug first) trades clean placebo estimation for reduced on-drug exposure.
- Addresses root cause; cost is typically longer trial or reduced carryover estimation ability.

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Design-level solutions address the root cause (carryover in the data) rather than the symptom (power loss in the analysis). Their cost is typically a longer trial duration or reduced ability to estimate the carryover dynamics themselves.

7.7 6.7 Qualitative comparison

Table~2 summarises the strategies along three dimensions: whether they discard observations, implementation complexity, and the expected direction of benefit based on the mechanistic account in Section 3.2. All entries in the 'Hypothesised benefit' column are qualitative; no simulation evidence is presented here, and each strategy should be evaluated empirically before use in trial planning.

A systematic simulation comparison of these strategies under Architecture B, analogous to the factorial comparison of analysis models in Section 3, would identify which approach best recovers the power lost to carryover for each trial design. We regard this comparison as important future work and note that the on-drug-only restriction and the weighted analysis are the most tractable candidates for a first empirical evaluation.

Table 2: Qualitative summary of analysis strategies for recovering power under Architecture B carryover. All benefit assessments are mechanistic hypotheses, not simulation-supported estimates.

Strategy	Discards data	Complexity	Hypothesised benefit
On-drug only	Yes	Low	Eliminates correlation erosion; forfeits off-drug reference (best for OL/Hybrid)
Weighted	No	Low	Reduces erosion damage proportionally; depends on weight calibration
Within-subject contrast	Aggregates	Low	Robust to carryover misspecification; less efficient than LME
Two-stage random slopes	No	Moderate	Unknown; separates estimation stages; robustness unverified
Exclude early off-drug	Some	Low	Removes most contaminated observations; improves contrast clarity
Design modification	N/A	N/A	Addresses root cause; cost is longer trial or reduced carryover estimation

8 7. Relationship to the Broader Literature

8.1 7.1 Treatment effect heterogeneity

[46] **The PATH Statement distinguishes risk modeling from effect modeling for predictive HTE analysis; this is an analysis-model distinction, not a DGP distinction.**

- Rothwell (2005) establishes the predictive/prognostic biomarker distinction; Kent et al. (2020) synthesize it into the PATH Statement.
- PATH's risk modeling approach stratifies on a multivariable risk score; effect modeling adds explicit interaction terms.
- PATH favors risk modeling and cautions against data-driven effect modeling because of low power, multiplicity, and overfitting.

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The statistical literature on treatment effect heterogeneity is extensive. Rothwell [@rothwell2005] sets out the foundational distinction between *predictive* biomarkers (those that modify the treatment effect) and *prognostic* biomarkers (those that predict outcome regardless of treatment). The current consensus statement on predictive HTE analysis is the Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement of Kent et al. [@kent2020path], which distinguishes two analytic strategies for identifying predictive biomarker effects in randomized trials: a *risk modeling* approach, in which a multivariable risk model (developed without a treatment term) is used to stratify patients and examine variation in absolute treatment benefit across risk strata, and an *effect modeling* approach, in which the regression includes explicit treatment-by-covariate interaction terms. PATH expresses caution about data-driven effect modeling on the grounds of low statistical power, multiplicity, and overfitting, and favors risk modeling when the application criteria are met.

[47] **PATH's analysis-model axis (risk vs. effect modeling) is orthogonal to our DGP axis; both PATH strategies implicitly assume Architecture A.**

- PATH's distinction is analysis-model; ours is DGP; they are orthogonal.
- Both PATH strategies assume heterogeneity is in the mean structure (Architecture A); Architecture B does not fit either PATH category.
- Architecture A = parametric interaction model; Architecture B = latent-class/mixture-model perspective.

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We note that the PATH risk-versus-effect-modeling distinction is an analysis-model axis. The Architecture A versus Architecture B distinction introduced in this paper is a data-generating-process axis, and the two are orthogonal. Both PATH analytic strategies implicitly assume that the underlying treatment effect heterogeneity is encoded in the mean structure of the outcome (the heterogeneity is identifiable from differences in conditional means), which corresponds to Architecture A as a DGP. Architecture B as a DGP is a separate phenomenon that does not map cleanly onto either of PATH's analysis-model categories: the interaction signal resides in the covariance structure rather than the mean, and the standard regression-based predictive HTE analysis methods endorsed by PATH may have reduced power under Architecture B precisely because they were developed under an implicit Architecture A assumption. Architecture A can accordingly be characterized as a *parametric interaction model* (in PATH's effect modeling sense). Architecture B can be characterized as a *latent class* or *mixture model* perspective in which the biomarker is a noisy indicator of class membership.

8.2 7.2 Prevalence of each architecture

[48] **Architecture A is the near-universal standard across the biomarker-moderated trial simulation literature; Architecture B is absent from all major threads except Hendrickson et al.**

- Simon (2010), Freidlin and Korn (2014), Renfro and Sargent (2017), Haller and Ulm (2018), and the platform trial literature all use Architecture A.
- N-of-1 simulation studies (1997, 2016, 2013) use hierarchical random-effects models, a variant of Architecture A.
- Wang and Schork (2019) and Schork (2022) share authorship with Hendrickson et al. yet adopt Architecture A, suggesting the differential-correlation choice is paper-specific.

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A survey of the simulation methodology literature reveals that Architecture A (direct mean moderation) is the near-universal standard. Simon [@simon2010] employs it throughout the simulation frameworks he develops for biomarker-stratified designs; Freidlin and Korn [@freidlin2014] adopt it for adaptive enrichment trials; Renfro and Sargent [@renfro2017] use it for basket and umbrella trials; and it underlies the platform trial literature throughout. Haller and Ulm [@haller2018], in simulation work on biomarker-by-treatment interaction estimation in randomized trials with prognostic covariates, likewise adopt the standard regression specification, focusing on power and bias of the interaction estimator under different covariate adjustment strategies. N-of-1 simulation studies, including Zucker et al. (1997), Araujo et al. [@araujo2016], and Duan et al. [@duan2013], use hierarchical random-effects models in which individual treatment effects are drawn from a population distribution, a variant of Architecture A in which the biomarker predicts the random treatment effect through the mean structure. Methodological work on power and design for N-of-1 trials with serial correlation and carryover, from both Wang and Schork

[@wang2019] and Schork [@schork2022], adopts the same Architecture A specification, even though this work shares senior authorship with Hendrickson et al. [@hendrickson2020]. It appears, then, that the differential- correlation parameterization in the latter is a paper-specific design choice rather than a methodological convention of any laboratory or research program.

[49] **Architecture B appears in latent variable / SEM / copula contexts but is unique in N-of-1 clinical trial simulation.**

- Architecture B appears in latent variable / SEM / Bayesian joint modeling / copula contexts (e.g., Rizopoulos 2012).
- In N-of-1 clinical trial simulation, Hendrickson et al. (2020) appears unique in using it.

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Architecture B (differential correlation via MVN) appears primarily in latent variable and structural equation modeling contexts, including the joint modeling framework discussed by Rizopoulos [@rizopoulos2012] (where treatment-state-dependent covariance relates longitudinal biomarkers to survival outcomes rather than to treatment response directly), Bayesian joint modeling frameworks, and copula-based simulation studies. In the clinical trial simulation literature specifically, the Hendrickson et al. (2020) framework appears to be unique in using differential correlation as the mechanism for biomarker-treatment interaction in an N-of-1 trial context.

[50] **The near-universal use of Architecture A means most published power estimates are optimistic for Architecture B biomarkers under carryover.**

- Virtually all published power estimates for biomarker-moderated trials assume Architecture A.
- These estimates may be optimistic for biomarkers that operate through Architecture B (covariance-structure mechanism).
- The additional carryover-induced power loss under Architecture B is not accounted for.

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This predominance of Architecture A has an important implication: the power estimates reported in the vast majority of biomarker-moderated trial simulation studies may be optimistic for biomarkers that operate through an Architecture B mechanism, because these studies do not account for the additional power loss that carryover produces when the interaction signal resides in the covariance structure rather than the mean structure.

8.3 7.3 Crossover and N-of-1 trial methodology

[51] **The crossover trial literature addresses carryover as a nuisance parameter but ignores the architectural distinction; the standard recommendation applies differently under each.**

- Senn (2002), Jones and Kenward (2014), and Sturdevant and Lumley (2021) treat carryover as a nuisance parameter; none engages with DGP architecture.
- Standard recommendation (include carryover in model or use adequate washout) applies to both architectures.
- Under Architecture B, crossover designs with short washout periods may be less suitable than previously recognized.

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The crossover trial literature, represented most thoroughly by Senn [@senn2002] and Jones and Kenward [@jones2014crossover], addresses carryover extensively but does not distinguish between architectures for biomarker-treatment interaction modeling. More recent methodological work by Sturdevant and Lumley [@sturdevant2021carryover] on testing for carryover effects via mixed-effects models likewise treats carryover as a nuisance parameter to be estimated and adjusted for, without engaging with how carryover interacts with biomarker-by-treatment effect signals. The standard recommendation, namely to always include carryover in the analysis model or to use adequate washout, applies to both architectures but has different power implications under each. The finding that Architecture B is more sensitive to carryover suggests that crossover designs with short washout periods may be less suitable for detecting correlation-based biomarker interactions than previously recognized.

8.4 7.4 Precision medicine trial design

[52] **Enrichment and biomarker-stratified designs assume Ar-**

chitecture A for power calculations; the architectural choice matters differently across three design families.

- Simon (2010) and Freidlin and Korn (2014) frame power calculations in the Architecture A regression idiom.
- Three design families warrant separate consideration: adaptive enrichment, treatment-switching, and basket/umbrella trials.
- The architectural assumption is rarely made explicit, so the practical implications are easy to overlook.

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Enrichment and biomarker-stratified designs, as discussed by Simon [@simon2010] and Freidlin and Korn [@freidlin2014], use Architecture A exclusively for power calculations. Three families of design are particularly worth considering in light of the architectural distinction.

[53] **Adaptive enrichment trials are vulnerable to Architecture B power shortfalls precisely when the biomarker mechanism is most uncertain.**

- Interim power calculations driving adaptive decisions use Architecture A; if Architecture B holds and washout occurs, realized power will be lower than predicted.
- The mismatch biases adaptation rules toward premature futility stopping or overly narrow enrichment criteria.
- The problem is worst when the biomarker mechanism is unknown, exactly when adaptive enrichment is most often proposed.

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Adaptive enrichment trials adjust the eligible population mid-trial based on emerging evidence of a biomarker-treatment interaction. The interim power calculations that drive these adaptive decisions are universally framed in the Architecture A regression idiom. If the true biomarker mechanism is closer to Architecture B and any participants pass through washout intervals between dose-finding stages, the realized power for the interaction test will be lower than the adaptation rule expects, biasing decisions toward premature futility stopping or toward overly narrow enrichment criteria. The discrepancy is most acute when the biomarker mechanism is uncertain at the design stage, which is precisely the situation in which adaptive enrichment is most attractive.

[54] **Treatment-switching designs suffer Architecture B power loss through washout erosion; parallel-group enrichment designs do not, because there is no off-drug phase.**

- Crossover, basket, platform, and N-of-1 hybrid designs all cycle participants through drug states; each washout phase erodes BM-BR correlation under Architecture B.
- Parallel-group enrichment designs escape this loss because each participant stays on a single treatment throughout.
- The architectural choice matters most in within-subject designs, where it is also most likely to be clinically important.

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Treatment-switching designs (most crossover, basket, and platform trials, as well as N-of-1 hybrid designs) cycle individual participants through different drug states. Under Architecture B, each washout phase erodes the BM-BR correlation, so the biomarker-treatment interaction signal in the second and subsequent treatment periods is structurally weaker than in the first. The power loss documented in Section 3 applies here directly. By contrast, parallel-group enrichment designs, in which each participant remains on a single treatment for the duration of the trial, escape this loss because there is no off-drug phase during which the correlation can decay; for these designs, Architectures A and B yield similar power estimates. The architectural choice therefore matters most in exactly the designs where the biomarker mechanism is most likely to be of clinical interest, namely those that deliberately expose each participant to both treatment states in order to estimate within-subject responsiveness.

[55] **Basket and umbrella trials face an additional risk of optimistic aggregated power estimates when Architecture B holds in even a subset of arms.**

- Renfro and Sargent (2017) discuss power calculations that average across biomarker-defined arms; the calculation assumes Architecture A throughout.
- Mixed architectures across baskets produce optimistic trial-level power estimates that are difficult to detect from aggregate results.
- Subgroup-specific power diagnostics under both architectures are

warranted when the biological mechanism is heterogeneous.

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Basket and umbrella trials, as discussed by Renfro and Sargent [@renfro2017], introduce an additional layer because their power calculations average across multiple biomarker-defined arms. If even a subset of the arms operate through an Architecture B mechanism while the trial-wide power calculation assumes Architecture A throughout, the aggregated power estimates will be biased optimistically in a way that is difficult to detect from trial-level results alone. Subgroup-specific power diagnostics under both architectures should accompany the primary planning analysis whenever the underlying biology is heterogeneous across baskets.

[56] **Across all three design families, the practical recommendation is to anchor sample size planning on the Architecture B estimate when the biomarker mechanism is uncertain.**

- Run power calculations under both architectures and treat Architecture B as the conservative anchor.
- This recommendation parallels the advice in Section 5.3 and is consistent with the PATH Statement (2020).
- The PATH Statement's caution about data-driven interaction discovery reinforces the case for conservative planning under architectural uncertainty.

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In all three settings the practical recommendation parallels that of Section 5.3: when the biological mechanism is uncertain, run the power calculation under both architectures and treat the Architecture B estimate as the more conservative anchor for sample size planning. This recommendation is consistent with the PATH Statement [@kent2020path], which itself counsels caution about data-driven discovery of treatment-by-covariate interactions and prefers risk modeling approaches when the application criteria are met.

9 8. Conclusions

In conclusion, six points are to be emphasized.

1. The choice between direct mean moderation (Architecture A) and differential correlation (Architecture B) for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.

2. Under Architecture A, carryover reduces power only modestly, by 1 to 3% in relative terms across the three designs examined, because the proportional biomarker-drug relationship is preserved during washout. Under Architecture B the loss is strongly design-dependent: it remains modest in the crossover and hybrid designs (3 to 5%) but reaches roughly 31% in the open-label OL+BDC design, because the differential-correlation signal erodes as the drug effect wanes.

3. The choice should be guided by the hypothesized biological mechanism: Architecture A for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics; Architecture B for biomarkers that statistically predict treatment response through latent subtype membership.

4. For the prazosin-PTSD application with blood pressure as the predictive biomarker, Architecture B (MVN differential correlation) is the more appropriate model, and the resulting power sensitivity to carryover should inform trial design decisions regarding off-drug period duration.

5. The existing clinical trial simulation literature uses Architecture A almost exclusively. The Hendrickson et al. (2020) MVN approach is, to the best of our knowledge, unique in the N-of-1 trial context. This means that published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based (Architecture B) mechanism, because they do not account for the carryover sensitivity documented here.

6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and consider sensitivity analyses under the alternative when the biological mechanism is uncertain.

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10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

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13 Reproducibility

This manuscript was rendered with R~4.6.0 inside the project's zzcollab Docker container (`rocker/tidyverse:4.6.0`; image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` with `mclapply` (Section~3.1 and 3.3 production runs), `furrr` with `future_multicore` (Architecture~C production run), `data.table` (the `original-extended` simulation collection), `purrr` (the `tidyverse` simulation collection), and `rmarkdown` for the manuscript build.

Sample-size convention. The Section~3.1 driver (`01-dgp-prototype.R`) generates $N = 35$ subjects independently per randomisation path, then concatenates paths before analysis; for two-path designs (CO, OL+BDC) the effective analysis sample is therefore 70 subjects, and for the four-path Hybrid design it is 140 subjects. The Architecture~C driver (`01-run-combined-factorial.R`) uses $N = 35$ as the total number of subjects, divided equally across paths, so that Section~3.4 figures refer to a 35-subject trial. The two parameterisations are not directly comparable; Section~3.4 power values are uniformly lower than their Section~3.1 counterparts for this reason.

Random-number generators. The Section~3.1 driver sets `set.seed(20260509)` and uses per-replicate seeds derived as `100000 * cell_index + rep_index`. The Architecture~C driver sets `set.seed(20260610)` and uses `furrr_options(seed = 20260610L)`.

Data and code availability. Sections 3.1 and 3.3: per-replicate outputs at `analysis/data/quick-sim/01-dgp-replicates.rds`; ADEMP cell-level summaries at `01-dgp-summary.txt`; driver at `analysis/scripts/quick-sim/01-dgp-prototype.R`.
Section~3.4 (Architecture~C): per-replicate outputs at `analysis/data/dgp-combined/01-combined-replicat`
cell-level summaries at `01-combined-summary.rds`; driver at `analysis/scripts/dgp-combined/01-run-combi`
The manuscript source, bibliography, and CSL file are at `analysis/report/01-dgp-mean-moderation-vs-mvn/`

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